

2(a)	(momentum =) mass $\times$ velocity	B1
2(b)(i)	time = 40 ms	A1
2(b)(ii)	1. (the magnitude of the acceleration is) constant	B1
	2. (the magnitude of the acceleration is) zero	B1
2(c)	$F = \Delta p / (\Delta)t$ or $F = \text{gradient}$	C1
	e.g. $F = 0.50 / 40 \times 10^{-3}$ $= 13 \text{ N}$	A1
2(d)	horizontal line from (0, 0.40) to (60, 0.40)	B1
	straight line from (60, 0.40) to (100, -0.10)	B1
	horizontal line from (100, -0.10) to (160, -0.10)	B1